

# Dmitri Lvovich

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## EDUCATION

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### McGill University

*Bachelor of Science, Major in Mathematics and Computer Science*

GPA: **3.72/4.00**

**Montreal, Canada**

*Aug 2023 – Apr 2027*

**Relevant Coursework:** Advanced Problem Solving in Competitive Programming, Algorithmic Game Theory, Advanced Algorithms, Applied Machine Learning

## WORK EXPERIENCE

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### Amazon – AWS Lambda

*Software Engineer Intern*

**Vancouver, Canada**

*May 2025 – Jul 2025*

- Designed and prototyped an open-source Apache Flink stream-processing solution for AWS Lambda's Event Source Mapping team, targeting high-throughput, large-scale event workloads
- Demonstrated a **20.3% increase** in throughput compared to the existing architecture

### Transport Canada

*Software Developer Intern*

**Ottawa, Canada**

*May 2024 – Aug 2024*

- Collaborated with clients to gather project requirements and delivered updates through regular demos
- Improved an office booking app used by 5000+ employees by resolving scheduling issues
- Enhanced internal tools with new features: Excel export-to-email functionality, notification system for permission group modifications, bulk permission assignment automation, and double-booking resolution plugin

## CONTESTS & AWARDS

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- **13<sup>th</sup> in Canada** in Round 2 of Meta Hacker Cup 2025
- 2023 Euclid Mathematics Contest School Champion
- 2022 Canadian Senior Mathematics Contest School Champion

## RESEARCH & PROJECTS

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### [Minecraft Structure Reconstruction](#) | *Research Project, Ongoing*

Developing a system that recovers the 3D block-level geometry of Minecraft structures from 2D input images

### MiniCaml Interpreter | *OCaml*

Implemented an interpreter for a functional language with capture-avoiding substitution, big-step evaluation, and type inference using unification

### Operating Systems Shell & Scheduler | *C*

- Built a Unix-like shell supporting built-in commands, command chaining, and fork/exec/wait
- Implemented a multi-process scheduler (FCFS, SJF, Round-Robin, Aging) using custom PCBs and ready queues
- Implemented virtual memory management with demand paging and an LRU page replacement policy

### Search Engine | *Java*

Implemented a search engine using web crawling and indexing; ranked pages based on relevance and user query

## SKILLS

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Competitive programming | C++ | C | Java | Python | OCaml | AWS